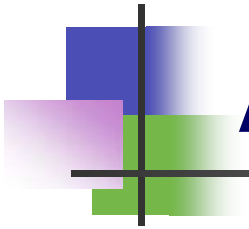


Chapter 24

Saturation Flow Rate



Announcements

- Tech memo 2 is due on February 22nd

Learning Objectives

- Establishment of lane groups
- Determination of saturation flow rates

Development of a traffic signal phasing and timing plan

- Determine phasing
- Establish analysis lane groups
- Calculate analysis flow rates and saturation flow rates
- Determine critical lane groups and lost times
- Calculate cycle length
- Allocate green time
- Calculate yellow & clearance intervals
- Check pedestrian crossing time

Source: HCM 2010

The adjusted saturation flow rate per lane is computed as follows.

$$s = s_o \, f_w \, f_{HV} \, f_g \, f_p \, f_{bb} \, f_a \, f_{LU} \, f_{LT} \, f_{RT} \, f_{Lpb} \, f_{Rpb}$$

where

- s = adjusted saturation flow rate (veh/h/ln),
- s_o = base saturation flow rate (pc/h/ln), → default: 1900 pc/h/ln
- f_w = adjustment factor for lane width,
- f_{HV} = adjustment factor for heavy vehicles in traffic stream,
- f_g = adjustment factor for approach grade,
- f_p = adjustment factor for existence of a parking lane and parking activity adjacent to lane group,
- f_{bb} = adjustment factor for blocking effect of local buses that stop within intersection area,
- f_a = adjustment factor for area type,
- f_{LU} = adjustment factor for lane utilization,
- f_{LT} = adjustment factor for left-turn vehicle presence in a lane group,
- f_{RT} = adjustment factor for right-turn vehicle presence in a lane group,
- f_{Lpb} = pedestrian adjustment factor for left-turn groups, and
- f_{Rpb} = pedestrian–bicycle adjustment factor for right-turn groups.

Source: HCM 2010

Adjustment for lane width

Average Lane Width (ft)	Adjustment Factor (f_w)
<10.0[a]	0.96
≥10.0–12.9	1.00
>12.9	1.04

Note: [a] Factors apply to average lane widths of 8.0 ft or more.

Source: HCM 2010

Adjustment for heavy vehicles

$$f_{HV} = \frac{100}{100 + P_{HV}(E_T - 1)}$$

where

P_{HV} = percent heavy vehicles in the corresponding movement group (%), and

E_T = equivalent number of through cars for each heavy vehicle = 2.0.

Source: HCM 2010

Adjustment for grade

$$f_g = 1 - \frac{P_g}{200}$$

where P_g is the approach grade for the corresponding movement group (%).

Source: HCM 2010

Adjustment for parking conditions

$$f_p = \frac{N - 0.1 - \frac{18 N_m}{3{,}600}}{N} \geq 0.050$$

where

N_m = parking maneuver rate adjacent to lane group (maneuvers/h), and

N = number of lanes in lane group (ln).

The parking maneuver rate corresponds to parking areas directly adjacent to the lane group and within 250 ft upstream of the stop line. A practical upper limit of 180 maneuvers/h should be maintained with Equation 18-8. A minimum value of f_p from this equation is 0.050. Each maneuver (either in or out) is assumed to block traffic in the lane next to the parking maneuver for an average of 18 s.

Source: HCM 2010

Adjustment for local bus blockage

$$f_{bb} = \frac{N - \frac{14.4 N_b}{3{,}600}}{N} \geq 0.050$$

where N is the number of lanes in lane group (ln) and N_b is the bus stopping rate on the subject approach (buses/h).

This factor should be used only when stopping buses block traffic flow in the subject lane group. A practical upper limit of 250 buses/h should be maintained with Equation 18-9. A minimum value of f_{bb} from this equation is 0.050. The factor used here assumes an average blockage time of 14.4 s during a green indication.

Source: HCM 2010

Adjustment for area type

The area type adjustment factor f_a accounts for the inefficiency of intersections in CBDs relative to those in other locations. When used, it has a value of 0.90.

Source: HCM 2010 and textbook

Adjustment for lane utilization

The input lane utilization adjustment factor is used to estimate saturation flow rate for a lane group with more than one exclusive lane. If the lane group has one shared lane or one exclusive lane, then this factor is 1.0.

Table 24.7: Default Values for the Lane Utilization Adjustment Factor

Lane Group Movements	No. of Lanes in Lane Group	Traffic in Most Heavily Traveled Lane (%)	Lane Utilization Factor f_{LU}
Through or Shared	1	100.0	1.000
	2	52.5	0.952
	3[a]	36.7	0.908
Exclusive Left Turn	1	100.0	1.000
	2[a]	51.5	0.971
Exclusive Right Turn	1	100.0	1.000
	2[a]	56.5	0.885

[a]If lane group has more lanes, field observations are recommended; if not, use smallest value shown in this exhibit.

(*Source:* Used with permission of Transportation Research Board, National Research Council, *Highway Capacity Manual*, 4th Edition, Washington DC, 2000, Exhibit 10-23, pp. 10–26.)

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Source: HCM 2010

Adjustment for right turns

The right-turn adjustment factor f_{RT} is intended primarily to reflect the effect of right-turn path geometry on saturation flow rate. The value of this adjustment factor is computed with Equation 18-10.

$$f_{RT} = \frac{1}{E_R}$$

where E_R is the equivalent number of through cars for a protected right-turning vehicle (= 1.18).

Source: HCM 2010

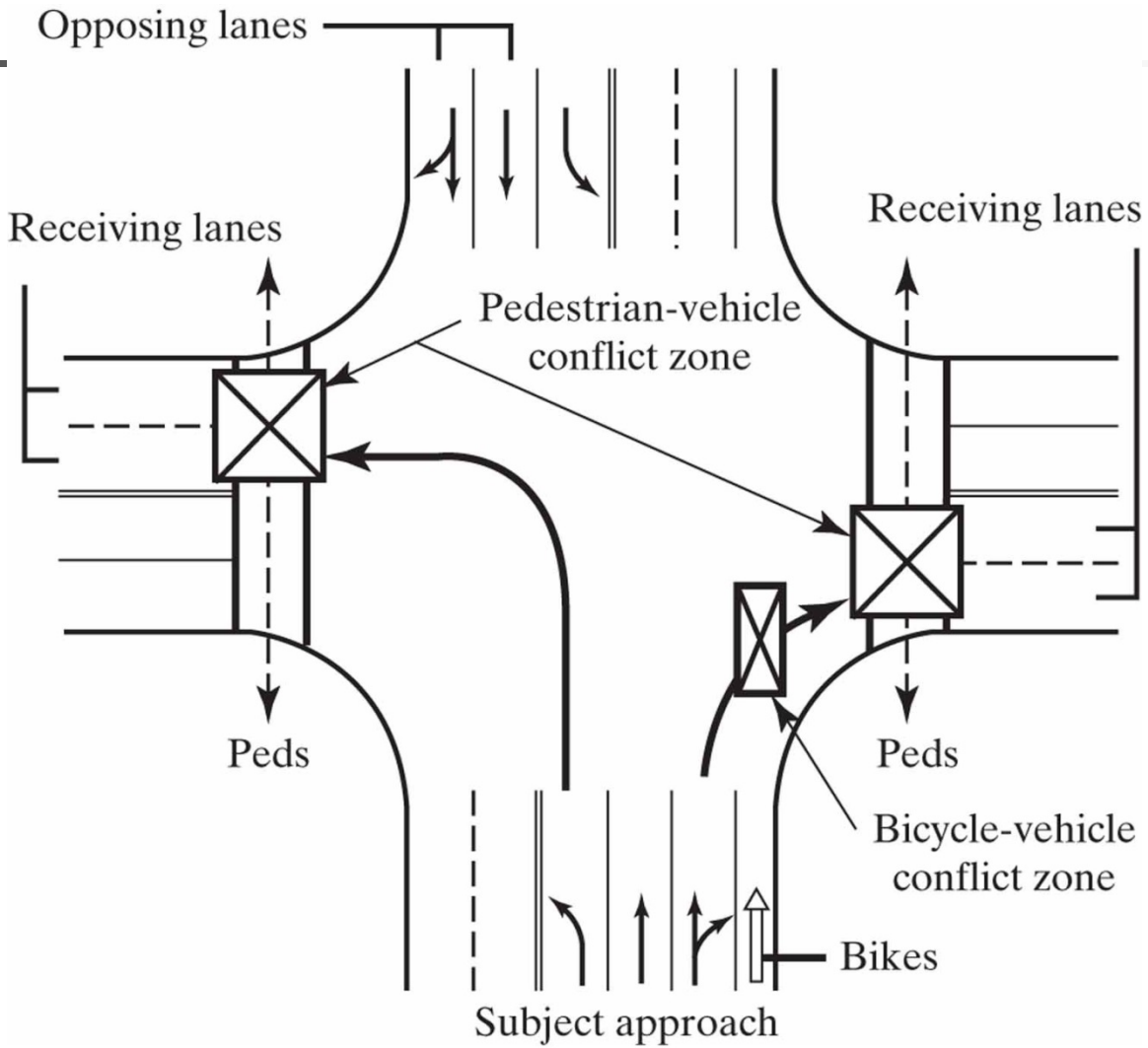
Adjustment for left turns

The left-turn adjustment factor f_{LT} is intended primarily to reflect the effect of left-turn path geometry on saturation flow rate. The value of this adjustment factor is computed with Equation 18-11.

$$f_{LT} = \frac{1}{E_L}$$

where E_L is the equivalent number of through cars for a protected left-turning vehicle (= 1.05).

Adjustment for pedestrians and bicycle interference with turning vehicles



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Pedestrian and Bike Interference Adjustment (continued):

Step 1: *Estimate Pedestrian Flow Rate During Green Phase (Left and Right Turns)* This is the actual pedestrian demand flow rate during the green phase. The rate is adjusted to reflect the fact that pedestrians are moving only during the green phase of the signal.

$$v_{pedg} = v_{ped}\left(\frac{C}{g_{ped}}\right) \leq 5{,}000 \qquad (24\text{-}17)$$

where: v_{pedg} = flow rate for pedestrians during the green phase, peds/hg

v_{ped} = demand flow rate for pedestrians during analysis period, peds/h

C = cycle length, s

g_{ped} = green phase for pedestrians, s

Step 2: *Estimate the Average Pedestrian Occupancy in the Conflict Zone (Left and Right Turns)* "Occupancy" measures represent the proportion of green time during which pedestrians and/or bicycles are present in a particular area for which the measure is defined. The occupancy of pedestrians in the conflict area of the crosswalk is given by Equation 24-18:

$$OCC_{pedg} = \frac{v_{pedg}}{2{,}000} \quad \text{for } v_{pedg} \leq 1{,}000$$

$$OCC_{pedg} = 0.40 + \left(\frac{v_{pedg}}{10{,}000}\right) \leq 0.90 \quad \text{for } 1{,}000 < v_{pedg} \leq 5{,}000 \qquad (24\text{-}18)$$

where: OCC_{pedg} = occupancy of the pedestrian conflict area by pedestrians

- 2,000 = 3,600/1.8 sec a pedestrian occupies the pedestrian conflict area.
- 10,000 = 3,600/0.36 sec a pedestrian occupies the pedestrian conflict area. 0.4 = 40% occupied.

Pedestrian and Bike Interference Adjustment (continued):

Step 3: *Estimate the Bicycle Flow Rate During the Green Phase (Right Turns Only)* The bicycle flow rate during the green phase is found in the same way that the pedestrian flow rate during green was estimated:

$$v_{bicg} = v_{bic}\left(\frac{C}{g}\right) \leq 1900 \qquad (24\text{-}19)$$

where: v_{bicg} = bicycle flow rate during the green phase, bic/hg

v_{bic} = demand flow rate for bicycles during the analysis period, bic/h

C = cycle length, s

g = effective green time for vehicular movement, s

Because bikes follow the same rule as cars

Step 4: *Estimate the Average Bicycle Occupancy in the Conflict Zone (Right Turns Only)* Bicycle occupancy may then be estimated as:

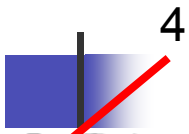
$$OCC_{bicg} = 0.02 + \left(\frac{v_{bicg}}{2700}\right) \qquad (24\text{-}20)$$

where: OCC_{bicg} = occupancy of the conflict area by bicycles

All other variables are as previously defined.

- 2,700 = 3,600/1.33 sec a bike occupies the bike conflict area.
- 0.02 = 2% occupied

Pedestrian and Bike Interference Adjustment (continued):



Step 5: *Estimate the Conflict Zone Occupancy (Left and Right Turns)* The occupancies computed in Steps 2 and 3 treat each element (pedestrians and bicycles) separately. Further, it is assumed that turning vehicles are present to block during all portions of the green phase.

Equation 24-21 is used to estimate the conflict zone occupancy for right turns from exclusive lanes with no bicycles present and for left turns from an exclusive lane on a one-way street:

$$OCC_r = \left(\frac{g_{ped}}{g}\right) OCC_{pedg} \qquad (24\text{-}21)$$

where: OCC_r = occupancy of the conflict zone

occupancy values would result in too great an adjustment. Allowing for the overlapping impact of pedestrians and bicycles on right-turning vehicles:

$$OCC_r = \left(\frac{g_{ped}}{g} OCC_{pedg}\right) + OCC_{bicg} - \left(\frac{g_{ped}}{g} OCC_{pedg} OCC_{bicg}\right) \qquad (24\text{-}22)$$

All terms are as previously defined.

Pedestrian and Bike Interference Adjustment (continued):

Step 6: *Estimate the Unblocked Portion of the Phase, A_{pbT} (Right and Left Turns)* Once the occupancies of the conflict zone have been established, the unblocked time for conflict zones (one for right turns, one for left turns) are computed as follows:

$$A_{pbT} = 1 - OCC_r \quad \text{if } N_{rec} = N_{turn}$$
$$A_{pbT} = 1 - 0.6OCC_r \quad \text{if } N_{rec} > N_{turn} \qquad (24\text{-}23)$$

where: N_{rec} = number of receiving lanes (lanes into which right- or left-turning movement are made)

N_{turn} = number of turning lanes (lanes from which right- or left-turning movement are made)

40% less impact

Pedestrian and Bike Interference Adjustment (continued):

Step 7: *Determine Adjustment Factors* The adjustment factors for interference of pedestrians and/or bicyclists with left-turn and right-turn movements can be determined as follows:

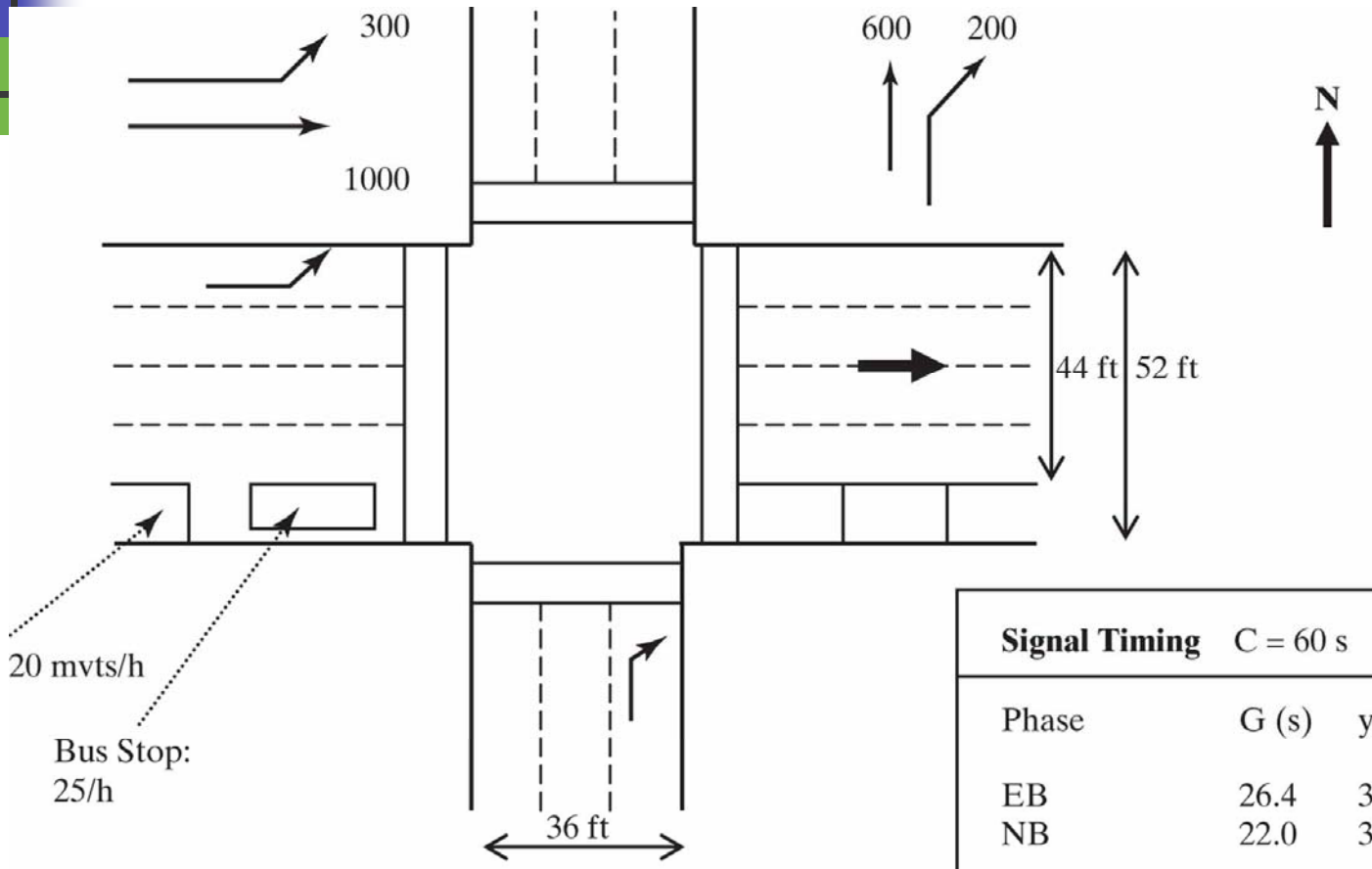
Adjustment Factor for Pedestrian and Bicycle Interference with Right Turns (f_{Rpb})

- If there are no conflicting pedestrians or bicyclists, $f_{Rpb} = 1.0$.
- If right turns are made as a protected phase, $f_{Rpb} = 1.0$.
- If right turns are made from an exclusive lane with permitted phasing, $f_{Rpb} = A_{pbT}$.
- If right turns are made from an exclusive lane with compound phasing, then $f_{Rpb} = A_{pbT}$ for the permitted portion of the phase and $f_{Rpb} = 1.0$ for the protected portion of the phase.
- If right turns are made from a shared lane, a complex procedure for determination of the shared lane-group saturation flow rate is presented in HCM 2010, Chapter 31.

Adjustment Factor for Pedestrian and Bicycle Interference with Left Turns (f_{Lpb})

- If there are no conflicting pedestrians, $f_{Lpb} = 1.0$.
- If left turns are made as a protected phase, $f_{Lpb} = 1.0$.
- If left turns are made from an exclusive lane on a one-way street with permitted phasing, $f_{Lpb} = A_{pbT}$.
- If left turns are made from an exclusive lane on a one-way street with compound phasing, then $f_{Lpb} = A_{pbT}$ for the permitted portion of the phase and $f_{Lpb} = 1.0$ for the protected portion of the phase.
- If left turns are made from a shared lane or on a two-way street, a complex procedure for determination of the shared lane-group saturation flow rate is presented in HCM 2010, Chapter 31.

Example



Signal Timing	C = 60 s		
Phase	G (s)	y (s)	ar (s)
EB	26.4	3.6	2.0
NB	22.0	3.6	2.4

HV% = 10% in all mvts.
100 peds/h in all crosswalks.
Crosswalk widths = 10 ft
Grades: NB +3%; EB level.

PHF = 0.90.
No RTOR.
Arrival Types: NB = 3; EB = 5.
Area Type: Fringe Area

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Calculation for flow rate for sample problem

Table 24.8: Flow Rate Computations for Sample Problem

Movement	*V* (veh/h)	*PHF*	Flow Rate (veh/h)	Lane Group Flow Rate (veh/h)
EB-LT	300	0.90	333	333
EB-TH	1000	0.90	1111	1111
NB-TH	600	0.90	667	667
NB-RT	200	0.90	222	222

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Estimation of saturation flow rates for sample problem

Table 24.11: Estimation of Saturation Flow Rates Using Equation 24-11

Lane Group	s_o (pc/hg/ln)	N (lanes)	f_w	f_{HV}	f_g	f_p	f_{bb}	f_a	f_{LU}	f_{RT}	f_{LT}	f_{Rpb}	f_{Lpb}	s (veh/hg)
EB LT	1900	1	1.000	0.909	1.000	1.000	1.000	1.000	1.000	1.000	0.952	1.000	0.932	**1533**
EB TH	1900	3	1.000	0.909	1.000	0.933	0.967	1.000	0.908	1.000	1.000	1.000	1.000	**4245**
NB TH	1900	2	1.000	0.909	0.985	1.000	1.000	1.000	0.952	1.000	1.000	1.000	1.000	**3240**
NB RT	1900	1	1.000	0.909	0.985	1.000	1.000	1.000	1.000	0.850	1.000	0.918	1.000	**1328**

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